VTX / VADER'S TAXIS · PHANTOM PROGRAMME

Phantom FX-A A10-B

Flight and engineering record

Investigation, test campaign and design decisions, 4 to 9 November 2026



Item	A10-B
Model	**phantom-a10b**, configuration **3ecae1a9**. The current, default Phantom in the simulator and the game (on main since 9 November 2026).
Derived from	A10-A as flown: phantom-fx with the P6 DRS stabilator, configuration 70089cca (66 of the 72 register runs).
Mass and power	1,933 kg. Four wheel fans, thrust-to-weight 1.38. Turbogenerator 600 kW continuous, 900 kW for 300 s, 80 kWh buffer.
Wing	Reference area 3.934 m², span 4.858 m, aspect ratio 6.0, incidence 7°.
Can (plant)	Drive and steer on the road; deploy the wings from a standing start; VTOL take-off and hover; fly, brake and turn with the fans facing down.
Cannot yet	Fly wing-borne with the fans off; rotate on a rolling take-off.

Evidence basis. Every result in this record comes from the MuJoCo physics plant driven by scripted test inputs. None are piloted game flights or measurements of a built vehicle. Each Phantom parameter carries a label: DERIVED, MANUFACTURER, REGULATORY_REFERENCE or SYNTHETIC_UNVALIDATED.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

Contents

§	Section
1	Summary
2	Stages of the work
3	Stage 1: the A10-A flight register and model identity
4	Stage 2: why the take-off roll does not rotate
5	Stage 3: hull lift and the Munk moment
6	Stage 4: design levers (canards, incidence, wing, battery and CG)
7	Stage 5: model fixes and the roll reruns
8	Stage 6: stability derivatives and road turning
9	Stage 7: real components and calibration
10	Stage 8: fan-assisted braking with the nose held
11	Stage 9: cornering and the 90° turn
12	Stage 10: wing deployment and dynamic pressure
13	Stage 11: VTOL and hover
14	Stage 12: mechanical front steering and the take-off lock
15	Stage 13: A10-B as the canonical model
16	Engineering decision log
17	A10-B characteristics
18	Open items and next steps
A	Test index and where the data is

1. Summary

The investigation started from the A10-A flight register: 72 recorded runs between 4 and 7 November, with a peak altitude of 600.58 m. A10-A hovered on its wheel fans, but no run achieved a conventional rolling take-off. The first questions were which model and physics those flights used, and whether that model could represent both hover and a horizontal take-off.

Each test was chosen from the telemetry of the one before. The work established five things:

- The rolling take-off cannot rotate. Lifting the nose about the rear tyres needs about 27 kN·m; the controls can supply 11 kN·m in the attitude law and 19 kN·m at full deflection. That holds at any speed the vehicle can reach.
- The plant was missing several physical terms: hull lift and its Munk moment, road suspension, the hub-motor power limit, steering and yaw control, a control law for wing-borne flight, and a dynamic-pressure check for the whole wing deployment.
- Some earlier "lift-offs" were not flight. They were tyre hops caused by a wheel overspinning when unloaded and then producing a contact-force spike when it touched down.
- Once the hull lift is modelled, the airframe is statically unstable in pitch ($C_{m\alpha}$ about +2.9/rad) and directionally unstable ($C_{n\beta}$ about −0.03/rad).
- With the fans facing down and a control law scheduled on dynamic pressure, the vehicle flies, brakes at up to 0.81 g with the nose held, and turns under control. With the fans off, it departs nose-down within about 2 s.

A10-B is the result. It is A10-A plus the following:

- four model fixes: hull lift, road suspension, wheel-drive limits and a wing-deployment guard;
- 7° wing incidence;

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

- control effectors calibrated to published hardware: front steer-by-wire, fin rudders, booster thrust vectoring, and Moog Model 965 actuators;
- a wing-borne control law.

A10-A remains reproducible under its original configuration identity. A10-B became the default Phantom and was merged into main on 9 November 2026.

Can (plant)	Cannot yet
Drive and steer on the road (18 m radius at 11 m/s)	Wing-borne flight with the fans off: departs nose-down within about 2 s at 90 m/s
Deploy the wings from a standing start, locked at 25.1 m/s (386 Pa)	Rolling take-off: rotation needs about 27 kN·m about the rear tyres; about 8–10 kN·m is available at 85 m/s
VTOL take-off and hover (thrust-to-weight 1.38)	
Brake from 95 m/s at up to 0.81 g with the nose held at about 4°, fans facing down	
Decelerating 90° turn ending in hover	

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

2. Stages of the work

The work ran as a chain. Each stage answered one question and, through its result, set the next one.

Stage	Question	Tests	Outcome
1	Which model and physics did the A10-A flights use?	Register audit, configuration hashing	70089cca (66 runs) and b835295b (4 runs) reproduced; hover and ground-roll physics present
2	Why does the take-off roll not rotate?	T1–T5	Moment budget about the rear tyres is about 3× short; flaps auto-retract at 75 m/s; tyre hops misread as lift-off
3	How much lift and pitching moment does the hull make?	M1–M6 (sections, Munk, vortex lattice)	dCm/dα +5.3/rad from the forebody; airframe becomes statically unstable
4	Which design levers help?	T3, T4 sensitivities	7° incidence adopted; canards and battery position insufficient on their own
5	What changes with hull lift and suspension?	R1–R5	Wheel overspin found and fixed with drive limits; top roll speed 92–99 m/s; no lift-off
6	Is it stable, and can it turn on the road?	U1, U2	Unstable in pitch and yaw with the hull term; no road steering existed
7	What real hardware can deliver the forces?	Component search	Moog Model 965 and ZF AKC adopted as calibration sources
8	Can it brake hard with the fans down and the nose held?	V1, V2	Hover-tuned law departs in 1.4 s; wing-borne law holds the nose, peak 0.81 g
9	What do the bank, momentum and arc of a 90° turn come to?	Calculation, V3a, V3b	Turn table; decelerating turn flown; fans-off flight departs
10	Can the wings fully deploy below 30 m/s?	W1, W1g	Yes from a standstill (locked at 25.1 m/s); the old sequencer could lock at 43.7 m/s; guard added
11	VTOL and hover	Capture runs	T/W 1.38; hover about 1.39 MW
12	Road steering and a take-off lock, made mechanical	Steering tests incl. zero-friction proof	Steer-by-wire on hinge joints; turns come from tyre forces
13	Make A10-B the canonical default	Regression tests, CI, merge checks	phantom-a10b default; A10-A identity pinned; merged to main

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

3. Stage 1: the A10-A flight register and model identity

The A10-A register lists 72 run IDs from 4 to 7 November 2026. Each run was matched to a parameter configuration by hashing the merged parameter tree, the same identity the simulator records.

- 66 runs used **70089cca**: phantom-fx with the P6 DRS stabilator switched on.
- 4 runs used **b835295b**: the same build with P6 off.
- 2 runs were on the 4 November Vercel sandbox.

The plant those runs used already contained the hover physics: four tilting wheel fans, governors, power bus, landing legs and ground contact. It also contained the ground-roll physics: tyres, hub drive, boosters, wing deployment and surface aerodynamics. It had no hull lift, no road suspension, no hub power limit and no road steering, and its fly-by-wire law was tuned for hover.

Recorded outcome	Runs
Airborne attempt ended	20
Collision	12
Landed	3
No take-off	33
Interrupted	1
Unclosed	2
No formal outcome	1

Peak altitude	Run
600.58 m	20261106T101512-085037
299.94 m	20261106T080610-01457b
74.96 m	20261104T023710-072007
73.28 m	20261105T212155-03a363

Decision. Keep A10-A reproducible. Every model fix was made an opt-in overlay, so the base configurations 70089cca and b835295b and every register grouping are unchanged. A test pins both identities. The fixes were gathered into a new variant rather than edited into A10-A.

4. Stage 2: why the take-off roll does not rotate

Test T1 instrumented a take-off roll and computed the pitching-moment budget about the rear tyre contact. Tests T2 to T5 followed from what T1 showed.

Test	What it showed
T1 instrumented roll	CG 1.438 m ahead of the rear tyre contact: weight moment 26.8 kN·m nose-down. Front tyres still carry 5.3 kN at 118 m/s. Pitch demand saturates at 0.209 (attitude gain 1.2/rad × 10° cap): canards +3.1°, stabilator 9.1°.
T2 per-surface aero	Wing CL 0.14 at zero incidence (2.2 kN per side at 115 m/s). Flaps retract automatically above 75 m/s. The stabilator at its 6° rest makes 1.37 kN of downforce, only 0.46 m behind the rear contact.
T3 static authority	Nose-up moment available at 120 m/s: 11.0 kN·m (attitude law), 18.9 kN·m (full stick) against 26.8 kN·m needed. With 7° incidence and full stick it would rotate at about 100–120 m/s. Tail-strike angle 11.9°.
T4 dynamic rolls	As flown: 0.07° pitch at most, to 115 m/s. With 7° incidence: rotates at 93.9 m/s, "wheels off" at 93.8 m/s, then tumbles. With 7° and the battery moved aft: wheels off at 92.7 m/s, then a collision.
T5 contact log	No hull-ground contact. At 94 m/s lift is 14.4 kN against an 18.6 kN weight share, so the "lift-off" was tyre hopping, with spikes to 76 kN. The road load path had no suspension: tyre only, 3.7 Hz, damping ratio 0.18.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

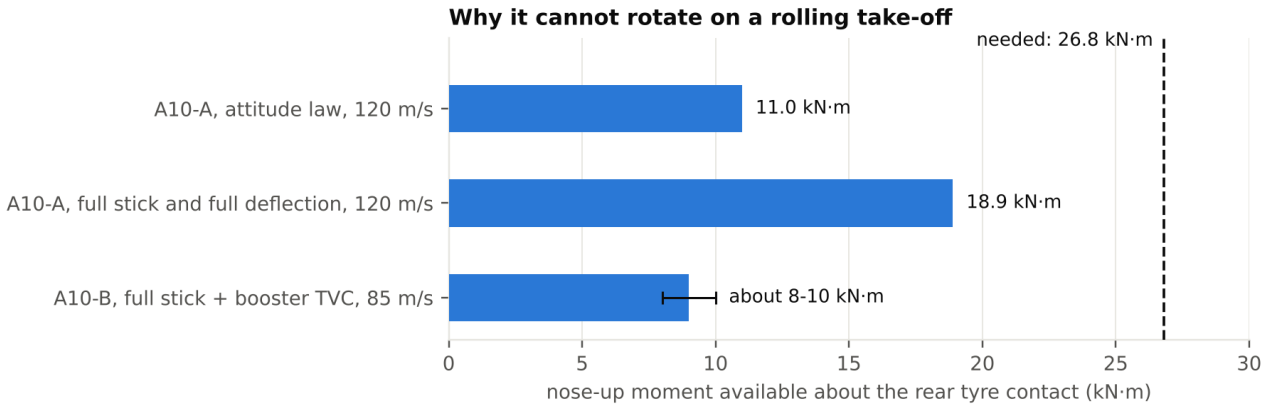


Chart 1. Nose-up moment available about the rear tyre contact against the moment needed. A10-A figures are from T3 at 120 m/s; the A10-B figure is from the take-off roll at 85 m/s, with full nose-up demand and booster thrust vectoring at −20°.

Several items in the model itself worked against the take-off, whatever the airframe design:

- the flaps auto-retract above 75 m/s, before any rotation speed;
- the attitude mode caps the pitch demand at 0.209;
- there was no road suspension;
- the wing zero-lift angle is hard-coded at −2°;
- there was no hull lift or Munk moment, and air density is held constant with altitude.

Decision. Treat the take-off as a moment-budget problem, not a gain-tuning problem. Raising controller gains cannot create moment the surfaces do not have. The model gaps were fixed first, and the take-off was re-tested on the corrected plant.



Figure 1. A10-B take-off roll at 84 m/s: rotation not achieved, with full nose-up demand and booster thrust vectoring at −20°. Two tyres are on the ground; dynamic pressure is 4,323 Pa.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

5. Stage 3: hull lift and the Munk moment

The hull is a broad, flat lifting body ahead of the CG, and the base plant gave it no lift. Exact plane sections were cut from the body-only CAD envelope. Its planform area is 10.38 m² against 10.14 m² in the parameter file, and its side area 4.07 m² against 3.96 m², both within 3 %. The pitching contribution was estimated four ways:

Estimate	dCm/dα (/rad, S 3.934 m², c 0.872 m)
Closed-body slender-body potential (upper bound)	+10.1
Multhopp forebody, no upwash credit	+5.7
Horseshoe vortex lattice, forebody (used)	+5.3 (K = 9.28 m² at x +1.81 m)
Vortex lattice, whole planform (check)	+5.2

With P6, the plant's static stability was Cm_α = −2.67/rad. With the hull term it is about +2.6/rad, which is statically unstable. The yaw counterpart, the side profile ahead of the fins, is K = 1.66 m² at x +1.30 m. It is destabilising against fins of about 1.06 m² at 1.91 m behind the CG.

Decision. Use the vortex-lattice forebody value (+5.3/rad), not the slender-body upper bound. It is computed on the actual CAD sections, and the independent Multhopp estimate agrees within 8 %. Only the forebody is added, because the wing model already spans the body. The term is written so it vanishes at 0° and 90° and holds in any flow direction.

6. Stage 4: design levers (canards, incidence, wing, battery and CG)

- **Canards.** At the saturated pitch demand the canards reach only +3.1°. They are too small and too close to the nose-down weight moment to lift the nose about the rear tyres on their own.
- **Wing incidence 7°.** This raises lift at a low body angle. In the A10-A static budget, 7° with full stick would rotate at about 100–120 m/s. In the dynamic rolls it brought the front tyres close to unloading at about 92–94 m/s, but no clean lift-off.
- **Wing size.** Unchanged in A10-B. With 7° incidence the wing carries about 0.8 of the weight at the 92–99 m/s the drive can reach. The rotation moment is about a third of what is needed, so a bigger wing alone does not fix the take-off.
- **Battery and CG.** Moving the 311 kg nose buffer from x +1.80 m to −0.06 m cuts the rotation moment to 21.1 kN·m. It also raises the hover load on the rear pods from 5.0 to 5.9 kN, against a 6.55 kN fan rating, which takes away most of the hover margin. The front crumple-zone extension and the battery layout were identified as the main forward-CG contributors.

Decision. Adopt 7° wing incidence (proposal P8) in A10-B. Keep the A10-A battery position, because hover margin matters more while VTOL is the working take-off mode. The forward-CG items (crumple zone, nose battery) are carried to the next airframe revision.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

7. Stage 5: model fixes and the roll reruns

Hull lift and a road suspension were added as overlays, and the roll series was rerun with the same schedule as T4: full stick from 70 m/s, then an attitude hold at 5° after the wheels leave the ground.

- **Hull lift** (hull_lift_v1): a forebody normal force at x +1.81 m (pitch) and a side-profile force at x +1.30 m (yaw).
- **Road suspension** (road_suspension_v1): one slide joint between each pod and its wheel. Ride frequency 1.6 Hz, damping ratio 0.35, preloaded to the design ride height, travel −0.12/+0.08 m. The tyre contact is tuned on the 19 kg unsprung wheel, because MuJoCo's contact stiffness is mass-normalised: on the 474 kg corner mass the tyre would come out about 25× too soft.

Run (R1: hull lift + suspension)	Result
As flown + suspension only	No rotation (0.82° max to 114 m/s); the suspension alone does not change the moment balance.
As flown + hull + suspension	Nose comes up at 106–107 m/s, then departs: pitch 2 → 4 → 12 → 42° in about 0.9 s with the controls saturated.
7° incidence + hull + suspension	Pitch held within ±1.3°; skimming touchdowns.
7° + battery aft + hull + suspension	Rotates from 81 m/s; wheels off at 82 m/s; then a 508°/s pitch-rate spike and a tumble.

Overspinning wheel (R2, R3)

The 508°/s spike was logged at 100 Hz and then step by step. The front-left wheel had unloaded at full droop while the full drive command was still applied. Nothing limited the hub motor, so it overspun to 452 rad/s, a tyre surface speed of 248 m/s at a road speed of 82 m/s. At touchdown, the soft contact removed that slip in one 2.5 ms step with a 208 kN force. The 76 kN spikes in the rigid-wheel T5 hops are probably the same mechanism.

The fix is **wheel_drive_limits_v1**. It caps the drive torque at P/ω using the hub motor's rated 60 kW (a parameter already in the file but unused by the base plant), and a slip limiter fades the torque out between 1× and 2× a slip ratio of 0.15.

R4 (all three fixes)	0–90 m/s	Speed at strip end	Wheels off
As flown	33.6 s, 1.94 km	99.4 m/s	never
7° incidence	37.1 s, 2.22 km	96.3 m/s	never
7° + battery aft	38.2 s, 2.32 km	92.1 m/s	never
7°, drive limits only (rigid wheels)	37.3 s, 2.24 km	96.0 m/s	never

- With the drive at its rated power, the roll levels off at 92–99 m/s. The base plant had run 600 N·m per wheel at any speed, about 104 kW per wheel at 95 m/s, so its acceleration figures were optimistic.
- The hops and "lift-offs" of T4, T5 and R1 do not occur once the wheels cannot overspin. They were a drive artifact, not lift.
- Wherever the nose does come up with the hull term on, the airframe departs in pitch within about a second.
- R5 re-flew the as-drawn A10 with the 4 November campaign's take-off schedule: no lift-off within the strip, with or without the drive limits. The 4 November "wheels off" results need re-checking against the drive limits.

Decision. Keep all three fixes (hull lift, suspension, drive limits) in A10-B. The drive limit uses a rating that was already in the parameter file, so it corrects the model rather than changing the design.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

8. Stage 6: stability derivatives and road turning

U1 swept the static pitch and yaw moments against angle of attack and sideslip, with and without the hull term. At 95 m/s with the wings out:

- Cm_α is +2.9/rad for A10-B (statically unstable) against −2.7/rad for A10-A;
- Cn_β is −0.03/rad (directionally unstable) against +0.10/rad for A10-A.

U2 tested turning on the road. The car could not turn: 0.0° of heading change with the yaw stick, and 0.6° by skid steering. The airframe had no road steering, and no yaw control in wing-borne flight.

Decision. Add front steering for the road, fin rudders for yaw in flight, and booster thrust vectoring for pitch and yaw authority that does not depend on airspeed. Size each from published hardware (Stage 7).

9. Stage 7: real components and calibration

Each new effector was sized from a published actuator, and its rate and torque limits were taken from the datasheet.

System	Component	Data used	Status
Canard, stabilator, rudder and TVC actuators	Moog Model 965 rotary servo actuator	249 / 282 N·m, 70°/s no-load at 28 VDC, ±90°, 5.2 kg	Adopted (MANUFACTURER)
Front steering actuator	ZF AKC 2nd generation class central actuator	11 kN rack force, for vehicles up to 3.5 t; speed not published	Adopted for force (MANUFACTURER)
Steering reference designs	Schaeffler Intelligent Corner Module; Hyundai Mobis e-Corner	Up to 90° wheel angle, steer-by-wire; no torque data	Reference
Steering rate reference	Chalmers rear-steer actuator study	At least 25°/s	Reference
Considered	Volz DA 58-D	80 N·m continuous, >200 N·m peak, 5.6 kg	Rejected: no published rate
Considered	Ultra Motion T-Series	12.2 N·m at 310°/s	Rejected: too little torque for the rudder at 120 m/s

The new systems add 35.6 kg to the mass ledger:

- five Moog actuators (two rudder, two TVC, one stabilator), 5.2 kg each;
- two rudder surfaces, 0.8 kg each (synthetic);
- the steering actuator, 8 kg (synthetic).

The canards moved from a synthetic 40°/s to the Moog 70°/s.

Decision. Calibrate to the Moog Model 965, because it publishes both torque and rate at a mass that fits. Size the steering from the ZF rack force: 11 kN × 0.12 m arm ÷ 2 = 660 N·m per front wheel. Where no figure is published, the value stays labelled SYNTHETIC_UNVALIDATED.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

10. Stage 8: fan-assisted braking with the nose held

The question was whether the vehicle could slow down quickly in controlled flight, using the wheel fans for stability, while the nose kept its attitude and the fans faced down.

- **V1**, the first airborne attempts, started untrimmed with 7° incidence. Both departed nose-down: −60° in 0.9 s on the aerodynamics alone, −63° in 1.1 s with the fans.
- **V2** used a trimmed start at 95 m/s with the nose at 4°. It compared the hover-tuned law with a new wing-borne law that schedules its gains on dynamic pressure.

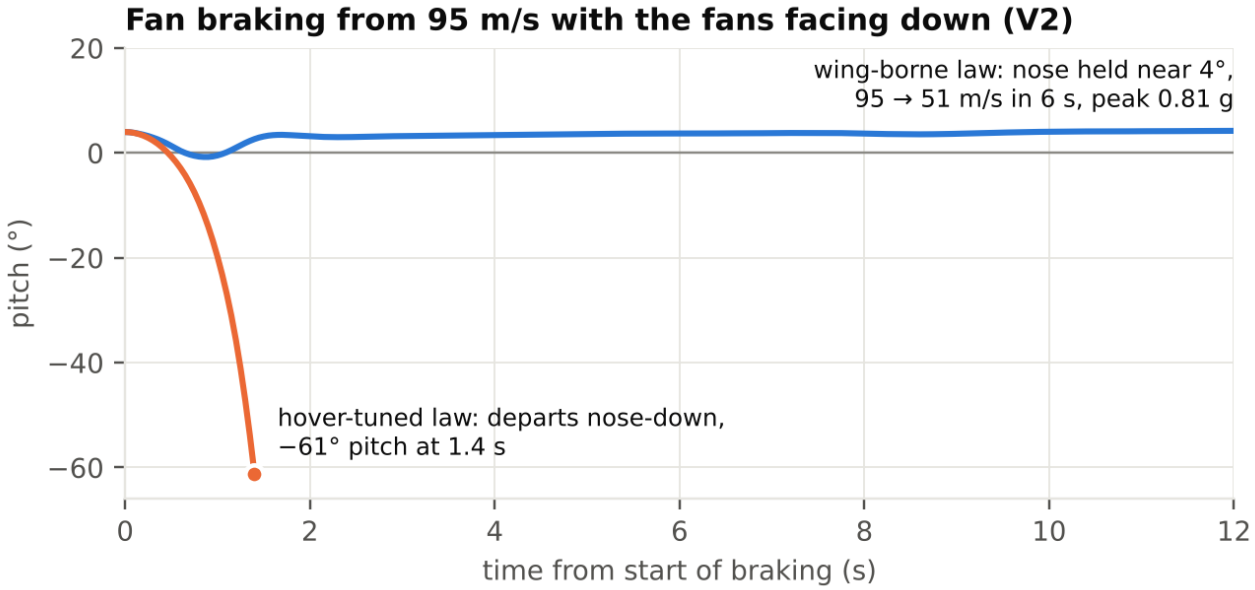


Chart 2. Pitch during fan braking from 95 m/s (V2). The hover-tuned law (attitude gain 1.2/rad, rate time constant 0.25 s) departs within 1.4 s. The wing-borne law (12/rad and 2.0 s in pitch, blended in between 400 and 2,000 Pa) holds the nose near 4°. It dips once, for 1.5 s, to −0.8°.

With the wing-borne law, V2 decelerated from 95 to 51 m/s in 6 s, with a peak of 0.81 g, altitude within 1.3 m and the nose held at about 4°. The fans' ram drag does most of the braking, and their differential thrust stays in the pitch loop.

Decision. Add the wing-borne control law, blended in on dynamic pressure, and keep the hover law below 400 Pa. Give the fast motion to the surfaces and thrust vectoring. Leave the fans, which lag about 0.3–0.9 s but are the strongest pitch effector at about ±14 kN·m, to hold trim and lift.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

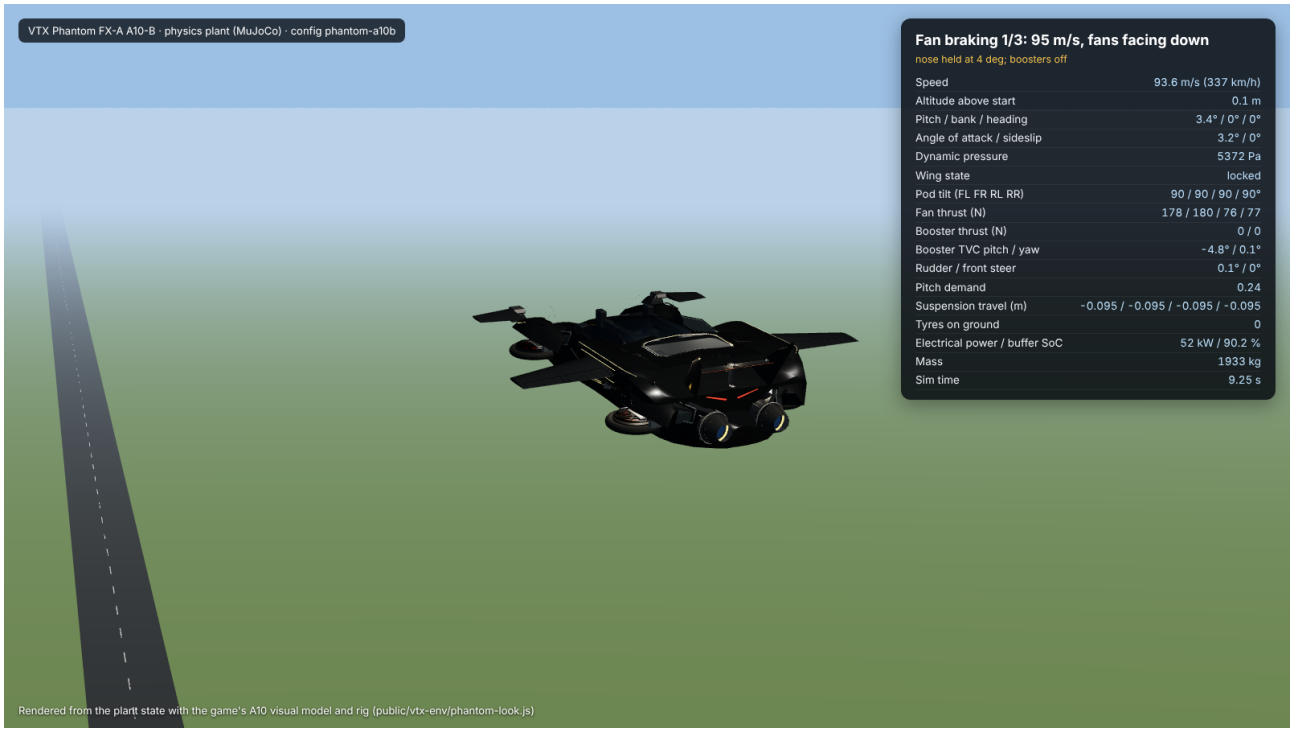


Figure 2. Fan braking 1/3: 95 m/s, fans facing down, nose held at 4°, boosters off.

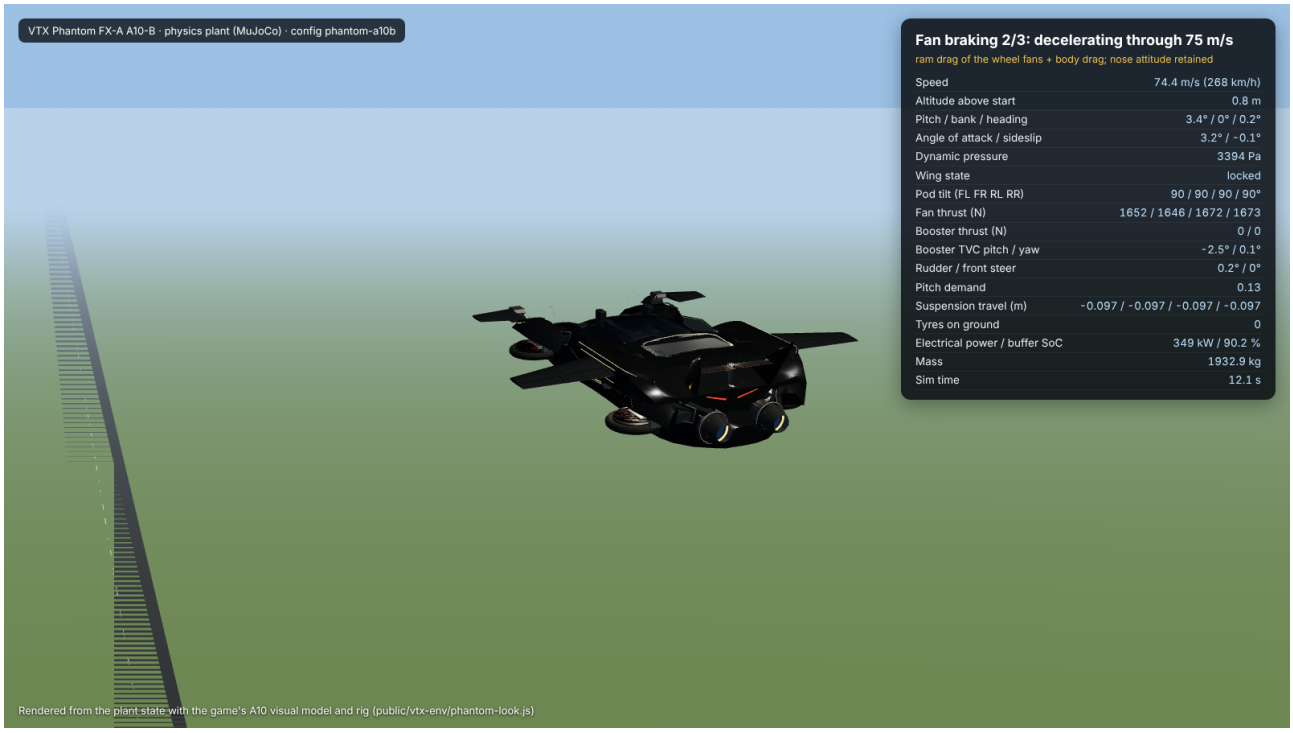


Figure 3. Fan braking 2/3: decelerating through 75 m/s on the wheel fans' ram drag plus body drag, with the nose attitude held.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

11. Stage 9: cornering and the 90° turn

First, the bank angle, radius, arc and yaw angular momentum of a coordinated, level 90° turn were calculated:

Speed	Bank	Radius	90° arc	Time	Yaw momentum (Izz·ω)
30 m/s	45°	92 m	144 m	4.8 s	1,630 N·m·s
45 m/s	30°	358 m	562 m	12.5 s	630 N·m·s
95 m/s	45°	920 m	1,446 m	15.2 s	520 N·m·s

Each fan rotor carries 251 N·m·s at hover speed. The four spin in opposing pairs, so the total cancels; the gyroscopic moment on each pod in these turns is 7–82 N·m. The front wheels also steer, but they cannot steer in flight: the take-off lock holds them within ±2° once the wings leave the hull (Stage 12).

V3b flew a decelerating 90° turn: braking from 95 m/s, a 30° bank from 60 m/s, and the nose held at about 4°. The turn is coordinated while there is airspeed. Below about 15 m/s the wing-borne law has blended out, and the nose lags the flight path. The track has turned 90° by about 14 s into the turn, at about 7 m/s, with the nose at −51°. The nose reaches 90° at 29.4 s, by yawing in near-hover while the vehicle drifts slowly sideways. So the 480 m and 29.4 s are measured to the 90° heading point and include that slow final segment.

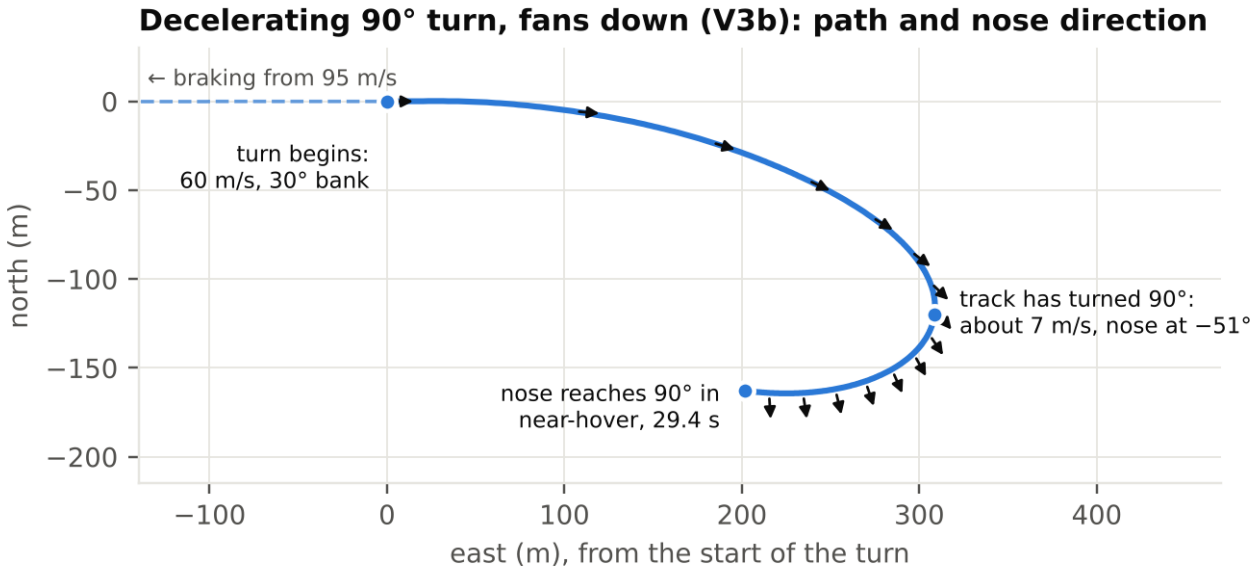


Chart 3. V3b ground track from the start of the turn, with the nose direction (arrows). The dashed segment is the end of the straight braking from 95 m/s. Altitude stayed within 1.3 m.

V3a tested wing-borne flight with the fans off at 90 m/s. The airframe departed nose-down within about 2 s, even with the surfaces and thrust vectoring saturated. The hull instability (Cm_α +2.9/rad) is larger than the aerodynamic control power.

Decision. For now, wing-borne flight in A10-B is fan-assisted: the fans stay in the pitch loop whenever collective is above 0.02. Flying with the fans off needs more pitch control power or less forebody lift (shaping), not more gain.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

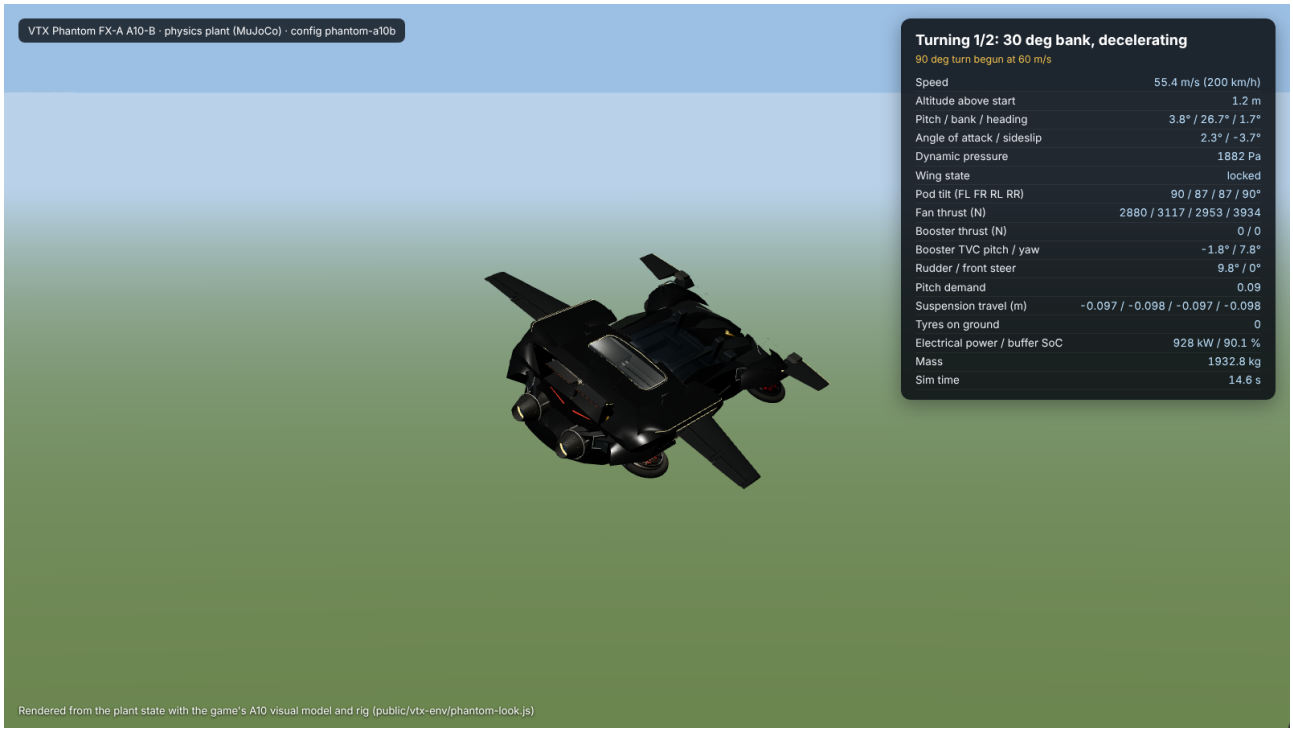


Figure 4. Turning 1/2: 30° bank, decelerating; the 90° turn began at 60 m/s.

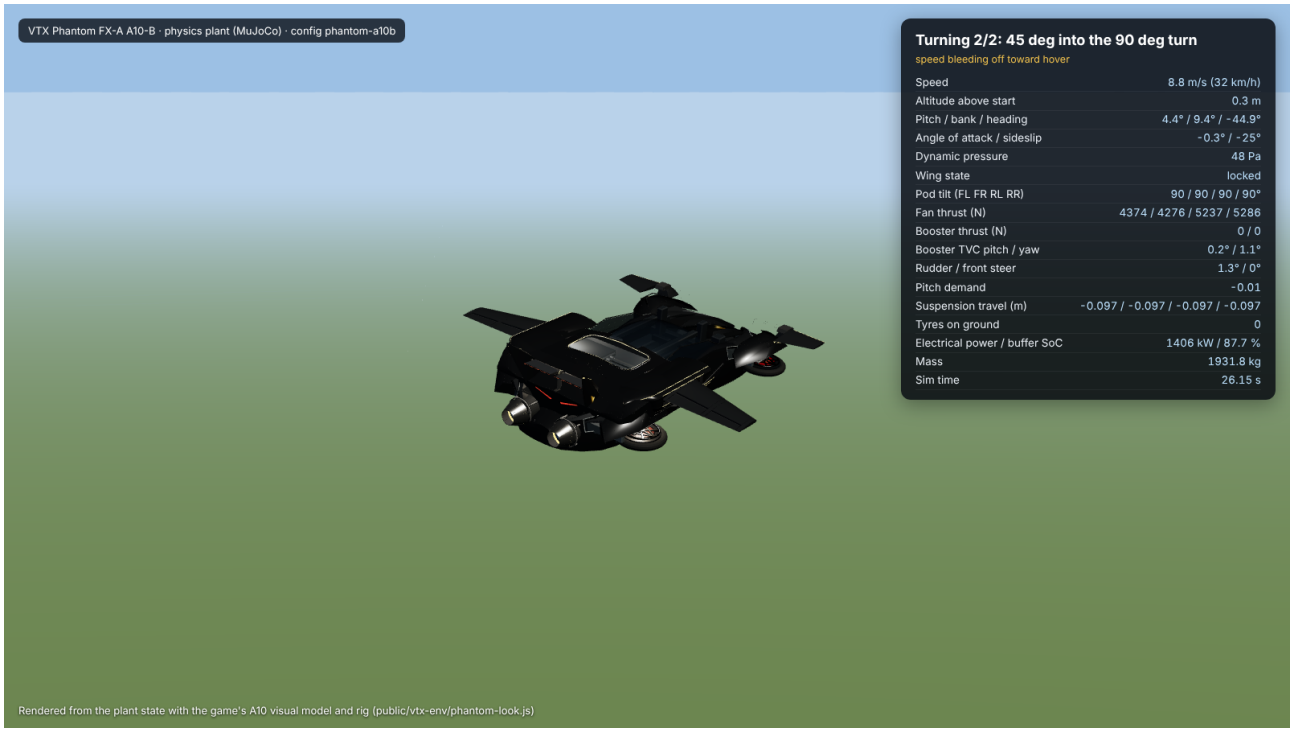


Figure 5. Turning 2/2: about 45° into the turn, speed bleeding off toward hover.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.



Figure 6. Turn complete: 90° of heading, ending in hover with altitude held.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

12. Stage 10: wing deployment and dynamic pressure

Requirement: the wings must fully deploy below 30 m/s, from a standing start with no brakes. The deployment sequence takes 5.52 s: hatch 1.0 s, slide 4.0 s, pins 0.5 s. The handover sets a transit limit of 600 Pa, which is 31.3 m/s.

Run	Result
Standstill, full drive and boosters (W1)	Locked at 25.1 m/s and 386 Pa. Peak in transit 383 Pa. Meets the requirement.
Old sequencer, started at 20 m/s while accelerating (W1)	Locked at 43.7 m/s and 1,171 Pa, nearly twice the limit. The sequencer checked dynamic pressure only when the slide started.
Old sequencer, started at 29 m/s (W1)	The hatch opened and the slide was then blocked, but the hatch stayed open as speed rose; it saw up to 6.36 kPa.
Guard, started at 20 m/s while accelerating (W1g)	Deployment inhibited: the predicted lock speed was above the limit.
Guard, started at 20 m/s while coasting (W1g)	Locked at 20.0 m/s and 244 Pa.

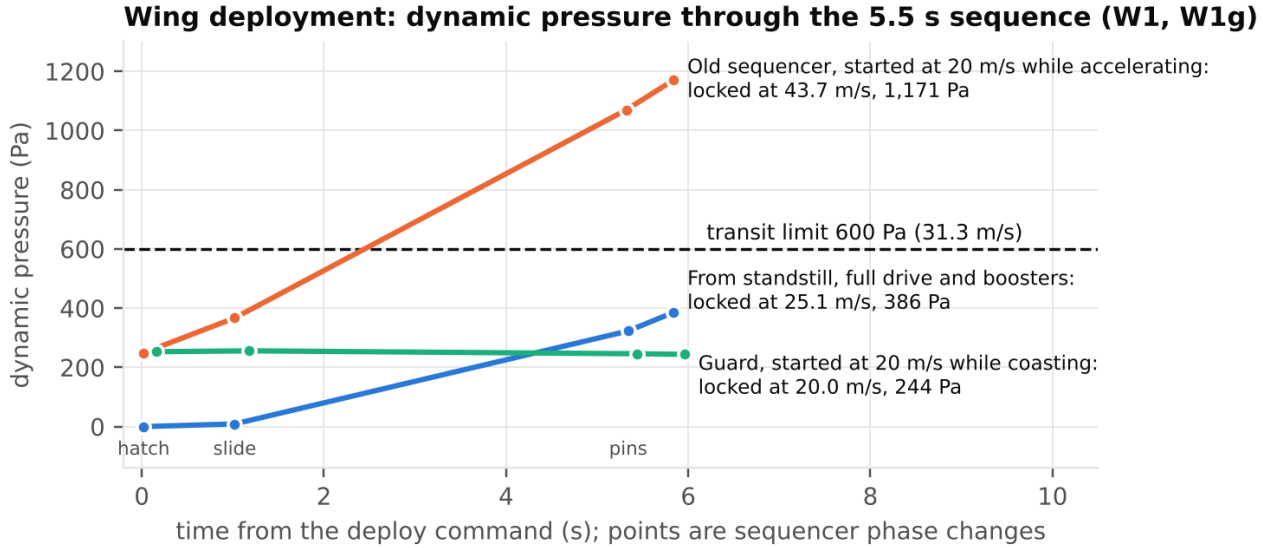


Chart 4. Dynamic pressure at each change of sequencer phase, measured from the deploy command. The lines join the phase points; they are not a continuous trace.

Decision. Add a deployment guard. Before starting, it predicts the lock speed from the current speed and an acceleration estimate over the 5.5 s sequence. It refuses to start if that would exceed the transit limit, and it closes the hatch if the limit is exceeded part-way. Each case is logged as an event.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.



Figure 7. Wing deployment 1/3: hatches opening; standing start, no brakes, full drive and boosters.

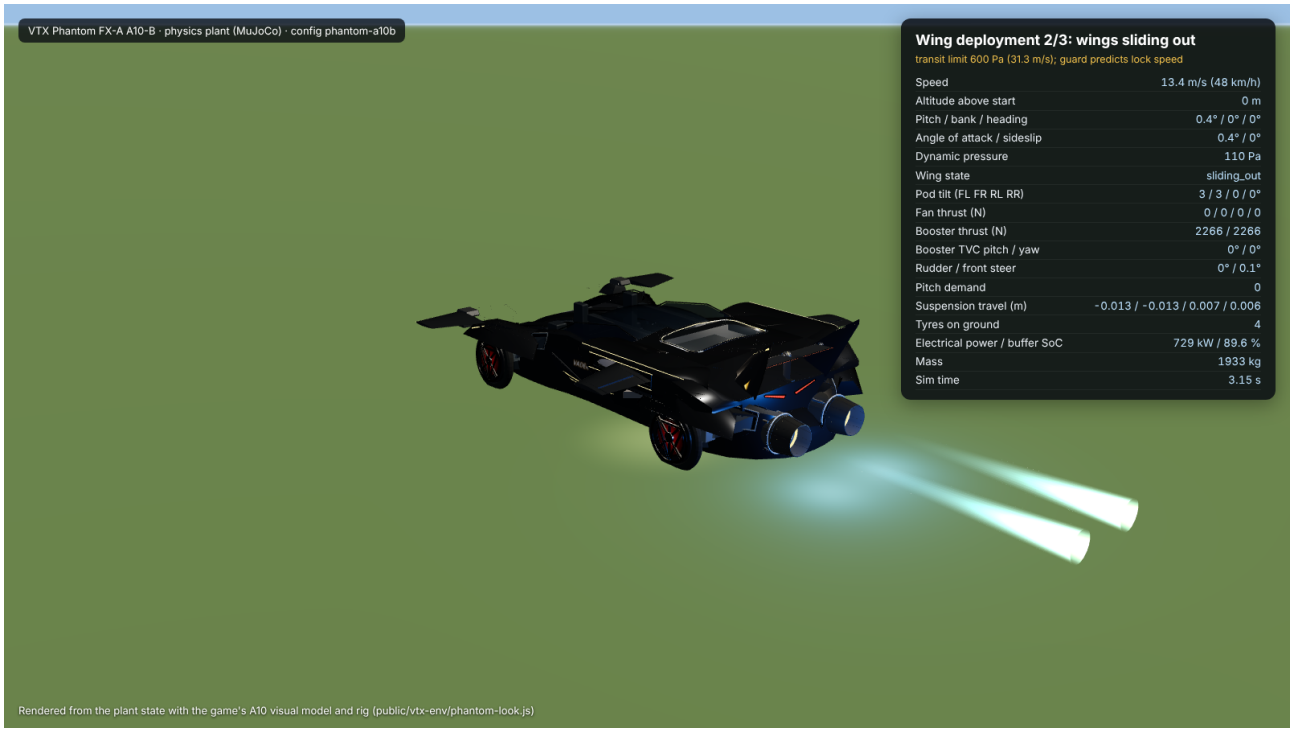


Figure 8. Wing deployment 2/3: wings sliding out at 13.4 m/s and 110 Pa.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.



Figure 9. Wing deployment 3/3: wings locked at about 25 m/s, inside the 600 Pa transit limit.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

13. Stage 11: VTOL and hover

With the pods rotated to 90° and the fans facing down, A10-B lifts off on its four wheel fans. Each fan gives 6.55 kN, for a thrust-to-weight ratio of 1.38. Hover takes about 1.3–1.4 MW: the 900 kW take-off rating, available for 300 s, plus the buffer. The front steering stays centred in VTOL.



Figure 10. VTOL take-off: lift-off on the wheel fans, pods at 90°, fans facing down.



Figure 11. VTOL hover at about 10 m: thrust-to-weight 1.38, hover power about 1.39 MW.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

14. Stage 12: mechanical front steering and the take-off lock

Steering had to be a physical mechanism, not a scripted heading change, and it had to fit the model's existing dynamics. The design:

- a hinge joint (vertical axis) on each front wheel, ahead of the suspension slide, driven by a torque-limited position servo: 660 N·m per wheel from the ZF rack force, critically damped at about 15 Hz;
- ±35° lock at 40°/s, from the yaw stick, only while the vehicle is on the road;
- the lock is scheduled with speed to keep lateral acceleration within 0.8 g;
- a take-off lock of ±2° at 10°/s once the wings leave the hull;
- centred in VTOL.

Proof that the steering is mechanical: a regression test sets the tyre and road friction to zero and applies the same steering input. The wheels steer, but the car does not turn (+0.1° against more than 15° with friction). The turn therefore comes from tyre forces. On the road the car turns on an 18.2 m radius at 11 m/s, against a kinematic 16.9 m, and changes heading by about 140° in 4 s.

Decision. Steer-by-wire on real joints, sized from published hardware and calibrated against the speed limit. The game rig animates the front pods from the plant's steer joints. Known issue: tyre-contact chatter at the 0.8 g limit; in one capture the number of tyres in contact drops to 2.

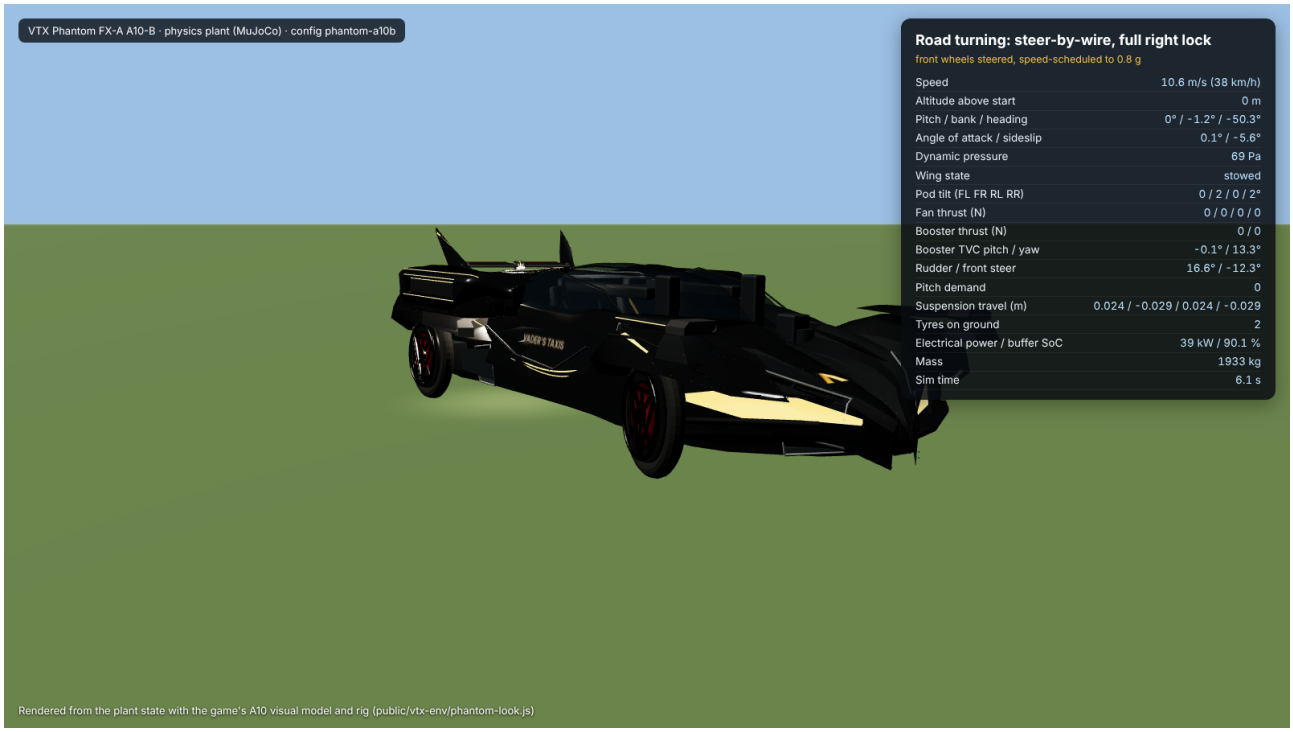


Figure 12. Road turning: steer-by-wire at full right lock, speed-scheduled to 0.8 g.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.



Figure 13. Road turning: 18 m radius at 11 m/s; heading change about 140° in 4 s.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

15. Stage 13: A10-B as the canonical model

- **Variant.** phantom-a10b is A10-A plus the following overlay files: hull_lift_v1, road_suspension_v1, wheel_drive_limits_v1 and control_effectors_v1. It also switches on P6 and P8 = 7°. Its configuration is 3ecae1a9, and it is the default Phantom (DEFAULT_VARIANT).
- **Previous model.** A10-A (phantom-fx) is labelled the previous prototype and keeps 70089cca and b835295b.
- **Game.** The vehicle list shows the Phantom first, with A10-B selected, and the cockpit opens on it. The API defaults stay R6 for the R6 pilot tooling.
- **Visual model.** public/assets/phantom-a10b-visual.glb carries the 7° wing incidence, and the front pods steer from the plant. The rudder surfaces are not in the mesh yet.
- **Tests.** There are 15 new regression tests (9 for A10-B, 6 for the overlays), covering:
  - identity: A10-B is the default and A10-A keeps its configurations;
  - composition and the added mass;
  - the steering: it is mechanical, centred in VTOL, and holds the take-off lock;
  - effector signs and the control-law blend;
  - the deployment guard.
- **Delivery.** The full suite passed (265 tests) and CI was green. A10-B was merged into main on 9 November 2026 (PR #25), with the README presenting VTX as a game.
- **Records.** The telemetry, logs, captures, images, scripts and the A10-A register are on the reference branch reference/phantom-a10b-20261109. A trial integration into the unified dashboard-cockpit candidate is kept on trial/a10b-into-dashboard-cockpit-20261109.



Figure 14. A10-B visual model, side view.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

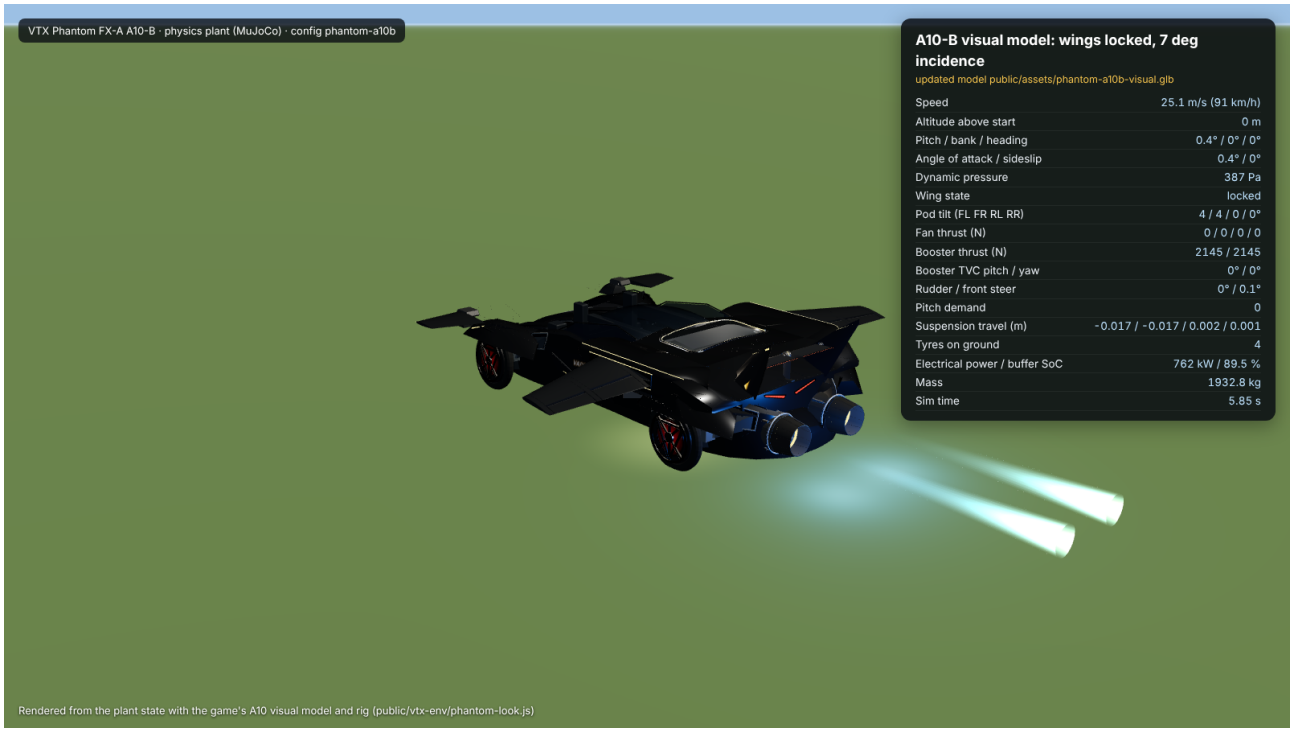


Figure 15. A10-B visual model, rear view.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

16. Engineering decision log

Decision	Reason	Alternatives considered
Model fixes as opt-in overlays; A10-B as a new variant	Keeps the A10-A register reproducible (70089cca, b835295b)	Editing the base parameters (would change every configuration identity)
Fix the model before tuning the take-off	The moment budget is 3× short; gains cannot create moment	Raising pitch gains or the attitude cap
Hull lift from the vortex lattice (+5.3/rad)	Uses the actual CAD sections; agrees with Multhopp within 8 %	Slender-body bound (+10.1/rad), too high for a flat, wide hull
Road suspension, tyre tuned on the unsprung mass	Rigid wheels hopped; MuJoCo contact stiffness is mass-normalised	Stiffer tyres only
Hub drive capped at the rated 60 kW plus a slip limiter	An unloaded wheel overspun and produced 208 kN contact spikes	Contact softening (would hide the cause)
Adopt 7° wing incidence (P8)	More lift at a low body angle; a small, well-understood change	Larger wing (does not fix the rotation moment)
Keep the battery position	Moving it aft costs most of the rear-pod hover margin	Battery at x −0.06 m (rotation moment 21.1 kN·m)
Wing-borne law scheduled on dynamic pressure	The hover-tuned law departed in 1.4 s	One gain set for all regimes
Rudders and booster TVC	Yaw control was missing; TVC gives authority independent of airspeed	Differential fan thrust only (slow)
Moog Model 965 for the surfaces and TVC	Published torque and rate at a suitable mass	Volz DA 58-D (no rate published); Ultra Motion (too weak)
Steer-by-wire on real hinge joints, from the ZF force	Steering must be physical; the turn must come from tyre forces	Scripted yaw (rejected outright)
±2° steering lock once the wings leave the hull	No wheel steering effects during take-off and flight	Free steering at all times
Wing-deployment guard with a lock-speed prediction	The old sequencer could lock at 1,171 Pa or leave the hatch open to 6.4 kPa	Check only at the start of the slide (the old behaviour)
Reference data on a separate branch	Keeps the canonical model branch clean for integration	Committing the data with the model

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

17. A10-B characteristics

Group	Values
Mass	1,933 kg; CG x −0.174 m, z 0.564 m (body frame); inertia Ixx 903, Iyy 4,280, Izz 4,990 kg·m²
Geometry	Length 4.89 m, body width 2.50 m, wheelbase 3.13 m, track 2.50 m, tyre radius 0.55 m
Wing	S 3.934 m², span 4.858 m, MAC 0.872 m, AR 6.0, aerodynamic centre x −0.806 m, incidence 7°, wing loading 4,819 Pa
Aerodynamics (95 m/s)	Cm_α +2.9/rad; Cn_β −0.03/rad. Free pitching moment −6.2 kN·m at 0°, −2.4 at 4°, +2.8 at 8°, +12.2 at 12°. Lift 12.7 kN at 0°, 23.2 kN at 4°
Propulsion	Four wheel fans, 6.55 kN each, 0.59 m² disk, T/W 1.38. Two EDF boosters, 2.57 kN static, 1.55 kN at 95 m/s. Hub motors 600 N·m, 60 kW
Power	Generator 600 kW continuous, 900 kW for 300 s; buffer 80 kWh at x +1.80 m (nose)
Chassis	Springs 42.3 kN/m front, 51.6 kN/m rear; dampers 2.9–3.6 kN·s/m; steering ±35° at 40°/s, 660 N·m per wheel
Effectors	Canards 70°/s; stabilator (P6); rudders ±25°, τ 0.55; booster TVC ±20° in pitch and yaw
Wing deployment	Sequence 5.52 s (hatch 1.0, slide 4.0, pins 0.5); transit limit 600 Pa (31.3 m/s)
Control law	Hover law below 400 Pa; wing-borne law blended to 2,000 Pa: pitch 12/rad, 2.0 s; roll 2/rad, 0.5 s; yaw 1.0 s; sideslip 3/rad; bank limit 45°

18. Open items and next steps

- **Rolling take-off.** Rotation needs about 3× the available pitch moment. Options: move the main gear toward the CG, raise the nose on its legs or suspension, or use a short take-off on the fans.
- **Wing-borne flight with the fans off.** Needs more pitch control power or less forebody lift through shaping, not more gain.
- **Not modelled yet:**
  - ducted-fan crossflow moments and loads, which in reality limit fan use at speed;
  - aerodynamic loads on the hatch, slide and pins during deployment;
  - ground effect on the hull;
  - density change with altitude.
- **Model settings to revisit:** flap auto-retraction above 75 m/s is still active, and the wing zero-lift angle is hard-coded at −2°.
- **Visual model.** The rudder surfaces and the TVC yaw deflection are not in the mesh yet.
- **Hard cornering.** Tyre-contact chatter at the 0.8 g limit.
- **Earlier results.** Re-check the 4 November campaign's "wheels off" results against the drive limits.
- **Integration with the unified environment.** Two changes are needed there. Its live server forces P6 on for every A10 variant, which relabels A10-B as configuration 71bb4a67; it should force P6 only where a variant lacks it. Its dashboard should preview the default variant at start-up.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.

Appendix A. Test index and where the data is

All files are on the branch reference/phantom-a10b-20261109, under reference/phantom_a10b_20261109/.

Test	Purpose	Data
T1	Instrumented take-off roll; moment budget about the rear tyre contact (A10-A)	telemetry/t1_P6.json
T2, T3	Per-surface aerodynamics; static nose-up authority	scripts/t2_aero.py, t3_static.py
T4	Dynamic rolls at full stick (as flown; 7°; 7° + battery aft)	telemetry/t4_*.json, logs/t4_*.out
T5	Ground-contact log of the 7° run (tyre hops)	telemetry/t5.json
M1–M6	Hull sections, CAD envelope, Munk and vortex-lattice estimates	telemetry/m2_sections.json, m4_cad.json, m5_munk.json
R1, R4	Rolls with hull lift, suspension and drive limits	telemetry/r1_*.json, logs/r1_*.out
R2, R3	100 Hz and per-step logs around the lift-off spike (wheel overspin)	telemetry/r2.json
R5	As-drawn A10 with the 4 November take-off schedule	logs/r5_base.out, r5_limited.out
U1	Static pitch and yaw moments against α and β (A10-A, with hull lift)	telemetry/u1.json
U2	Road turning before steering existed	scripts/u2_road.py
V1, V2	Airborne with the fans down; braking with the old and new laws	telemetry/v1_*.json, v2_*.json
V3	Fans-off wing-borne turn; fan-assisted decelerating turn; take-off roll	telemetry/v3_*.json
W1, W1g	Wing deployment without and with the guard	logs/w1_*.out, w1g_*.out
Captures	Plant snapshots behind the 15 figures	captures/*.json, images/*.png
Register	A10-A flight register, 4–7 November 2026	register/A10A_flight_register_20261108.json and .pdf

On main: the specification docs/phantom/PHANTOM_A10B_MODEL.md, the characteristics evidence/phantom/a10b_characteristics.json, the hull-lift calculation evidence/phantom/hull_lift_vlm.json, the overlays inputs/phantom_overlays/*.json, and the tests tests/test_phantom_a10b.py and tests/test_phantom_overlays.py.

Simulation evidence only (MuJoCo physics plant). © 2026 Steven Dark (SDarkVader). All rights reserved.